

Lab 1: Knowledge Is Power (KIP) Description Paper

Christopher Humphreys

CS 411W

Instructor: Thomas J. Kennedy

Submission Date: September 29, 2026

Version 1

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1 Introduction

Knowledge Is Power, or KIP, is a web based legal literacy platform designed to help young adults understand laws that affect everyday decisions. The project addresses gaps in practical legal education by combining short learning activities, realistic scenarios, clear feedback, and source based explanations. KIP is intended to give users a place to practice recognizing legal issues before they face those situations in real life. The prototype demonstrates this approach through a browser based system with learning modules, games, progress tracking, and recommendations based on user performance.

1.1 Societal Problem

Young people ages 16 to 19 begin taking on responsibilities involving driving, digital conduct, work, school, ownership, contracts, and personal safety. These responsibilities can have legal consequences, but practical law related education is not consistently included in K 12 instruction. The Touro Law Jacob D. Fuchsberg Law Center (2017) discusses the value of law related education for younger students and the importance of connecting legal concepts to situations they may actually encounter.

Driving shows why this gap matters. The Centers for Disease Control and Prevention (2026) and the National Highway Traffic Safety Administration (2024) identify teen drivers as a group with increased safety risks and emphasize informed decision making. The Legal Services Corporation (2017) also documents broader difficulty with civil legal needs. Without a reliable foundation, young adults may depend on friends, family, social media, or incomplete online information when trying to understand the law.

1.2 Needed Solution Characteristics

An effective solution should do more than present definitions for users to memorize. It should use realistic situations that require users to apply legal concepts, make choices, and receive clear explanations. The material should use consistent language, remain accessible on common devices, and allow students to complete low stakes practice independently.

The solution should also connect lessons to official legal sources and introduce legal vocabulary in context. Progress tracking can identify areas where a user needs more practice, while recommendations can direct that user toward relevant content without preventing free exploration of the platform.

1.3 Product Introduction

KIP is designed to meet these needs through interactive legal learning modules and games. Users can challenge misconceptions, work through realistic scenarios, review the consequences of their choices, and see the legal reasoning behind the feedback they receive. The platform records progress and uses performance information to recommend future practice. It is intended as a supplemental educational resource for young adults, parents, teachers, and school administrators.

2 KIP (Knowledge Is Power) Product Description

KIP is a web based educational service built around learning modules, interactive games, user progress data, and legal content connected to official sources. The platform uses short misconception based questions and longer branching scenarios to help users practice legal reasoning. After a user makes a choice, KIP provides feedback that explains the result and connects the lesson to the legal information behind it. User performance is stored so the system can recommend future games, modules, and scenarios that address areas where more practice

may be useful. The overall goal is to assess, improve, and reinforce practical legal knowledge through repeated practice that users can complete independently.

2.1 Key Product Features and Capabilities

The main features of KIP are designed to work together rather than function as separate educational tools. The games provide practice, the feedback explains the legal reasoning, the source reference system shows where the information comes from, and progress tracking records what the user has completed and how well the user performed. These features allow the platform to move beyond passive reading and give users repeated opportunities to apply legal concepts to decisions.

2.1.1 Solution Overview

KIP uses focused games, realistic examples, curated media, and official legal resources to teach practical legal boundaries. Users can browse available learning modules and games, or they can follow recommendations generated from their previous progress. This structure allows users to explore topics freely while still giving the system a way to guide them toward areas that need more attention.

2.1.2 Goals and Objectives

The primary goal of KIP is to improve legal literacy among young adults by making legal information easier to recognize and apply. The platform is designed to assess a user's current understanding, reinforce correct knowledge, identify misconceptions, and provide additional practice when needed. The user experience is intended to support recognition of legal situations, stronger decision making, and long term retention of important concepts without requiring a classroom setting.

2.1.3 Interactive Gamification

KIP uses low stakes games to make practice more active and approachable. The games are not intended to replace formal instruction or professional legal advice. Instead, they provide a structured environment where users can make choices, receive explanations, and see how context can change the legal significance of a situation.

2.1.3.1 Fact or Fiction? Game

Fact or Fiction? is a short form game built around common legal beliefs and misconceptions. A user is presented with a statement and chooses whether it is true, false, or dependent on context. The explanation that follows helps the user identify assumptions and recognize that some legal questions cannot be answered correctly without additional facts. The game is intended to build foundational knowledge before users move into longer scenarios.

2.1.3.2 Choose the Outcome Game

Choose the Outcome is a longer multiple choice game that presents a realistic situation and allows the user to examine people, objects, or documents before making a decision. Each choice leads to a consequence and an explanation of its legal significance. This format encourages users to slow down, consider the available information, and recognize which facts can affect an outcome.

2.1.4 Progress Tracking and Analytics

KIP tracks completion, topical strengths, and topical weaknesses during use. This information supports recommendations for games and modules and can also influence the scenarios selected for a game session. Content that tests weaker areas can be prioritized, while recently completed material can be reduced when possible. The purpose of this tracking is to

keep practice relevant to the user instead of presenting the same material in the same order for everyone.

2.2 Major Components (Hardware/Software)

KIP is organized into presentation, application, data, and hardware layers. The Major Functional Component Diagram shows how the user facing components communicate with the application systems and databases. Separating these responsibilities allows the interface, program logic, and stored data to be developed and maintained as distinct parts of the system.

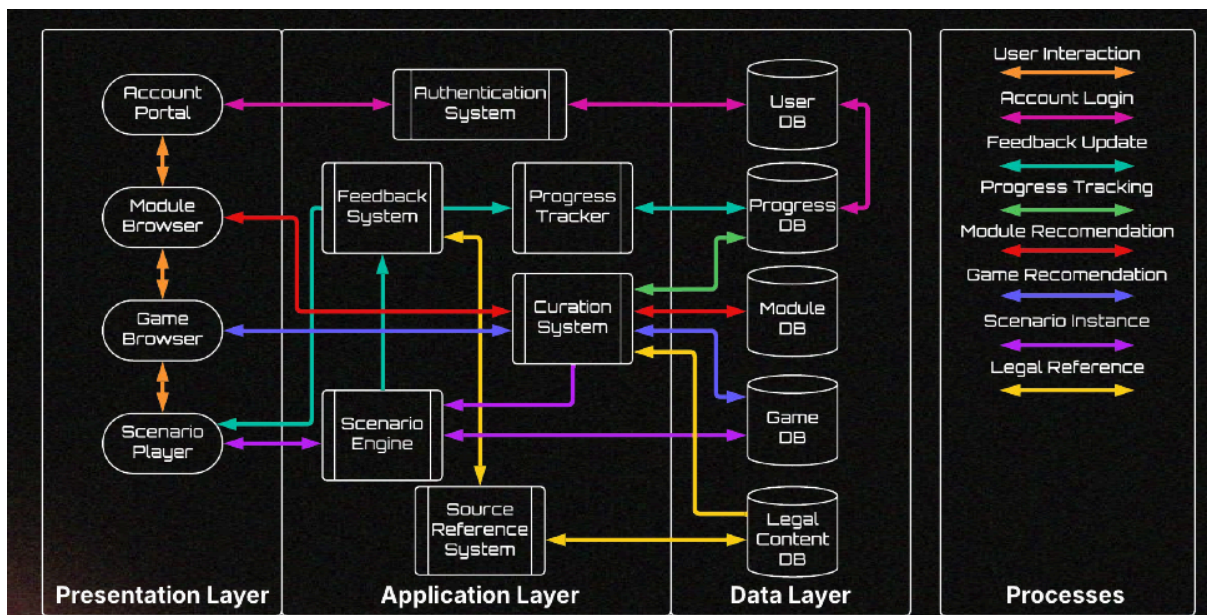


Figure 1. KIP Major Functional Component Diagram (MFCD)

2.2.1 Presentation Layer

The presentation layer is the part of KIP that users interact with directly. It uses JavaScript, Bootstrap, and HTML5 to provide the account portal, module browser, game browser, and scenario player. The account portal supports registered accounts and limited guest access. The module browser displays legal learning topics and progress, while the game browser

displays available games and recommendations. During gameplay, the scenario player presents questions, choices, consequences, feedback, and source references.

Table 1. Presentation Layer

Component	Description
Tech Stack	JavaScript, Bootstrap, and HTML5.
Account Portal	Supports login, account creation, guest access, and access to features based on account status.
Module Browser	Displays available legal learning modules and the user progress associated with each module.
Game Browser	Displays available games and prioritizes recommendations using progress and performance history.
Scenario Player	Presents interactive questions, branching scenarios, feedback, consequences, and source references during gameplay.

2.2.2 Application Layer

The application layer contains the program logic that supports KIP. Java and JDBC are used to manage the systems that connect user actions to stored data and learning content. The authentication system handles login, account creation, guest access, and session validation. The scenario engine loads the content used during games and manages the sequence of choices, branches, consequences, and game states. The feedback system explains the result of a user decision, while the curation system decides which games, modules, and scenarios should be prioritized. The source reference system connects educational content to legal sources and supports maintenance when a referenced law or resource changes.

Table 2. Application Layer

Component	Description
Tech Stack	Java and JDBC.
Authentication System	Manages login, account creation, guest access, and session validation.
Scenario Engine	Loads game content and manages choices, branches, consequences, and game states.
Feedback System	Explains results and the legal reasoning behind each decision.
Curation System	Recommends modules, games, and scenarios using progress and performance history.
Source Reference System	Links content to official legal sources and flags items for review when sources change.

2.2.3 Data Layer

The data layer stores the information KIP needs across sessions. SQLite is used for persistent storage. The planned data includes user account information, progress records, learning modules, game content, scenario information, and legal and educational resources. Although the architecture identifies separate databases by responsibility, these areas can be implemented as related structures within the project database while preserving the same functional separation.

Table 3. Data Layer

Component	Description
Tech Stack	SQLite.
Databases	User Database, Progress Database, Module Database, Game Database, and Legal Content Database.
Functions	Stores user information, progress, learning modules, game and scenario content, and legal and educational resources.

2.2.4 Hardware Layer

The hardware layer includes the server resources used to host KIP and the consumer devices used to access it. A virtual private server hosts the Java application, performs server processing, and provides access to the SQLite database file and application resources. Users access KIP from devices with a modern web browser and an internet connection. This arrangement supports a web based service that does not require users to install special software or use dedicated hardware.

Table 4. Hardware Layer

Component	Description
Server Side Components	A virtual private server that hosts the Java application, performs server processing, and provides access to data and application resources.
Client Side Components	Consumer devices with a modern web browser and an internet connection.
Functions	Runs application code, stores persistent data, processes requests, and delivers the web interface to users.

4 Glossary

CDC: Centers for Disease Control and Prevention.

KIP: Knowledge Is Power.

MFCD: Major Functional Component Diagram.

UX: User Experience.

5 References

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